FIFA Player Dataset Analysis Examples

1- Frequency Table (Absolute / Raw Counts)

Q: How many players does each league have in the dataset?
س :كم عدد اللاعبين الذين يمثل كل دوري في قاعدة البيانات؟
Tool: Count each league's players in a frequency table. Look for which league dominates the dataset — and whether the league with the most players also has the highest mean Overall Rating.

2- Frequency Table (Relative / Percentages)

Q: What share of all players does each position group (Forward, Midfielder, Defender, Goalkeeper) represent?
س :ما النسبة المئوية التي تمثلها كل مجموعة مراكز)مهاجمون، وسط، مدافعون، حراس (من إجمالي اللاعبين؟
Tool: Divide each position group's count by 18,207 to get percentages. Look for whether the distribution is roughly balanced, and whether the rarest position group commands the highest mean market value.

3- Frequency Table (Cumulative)

Q: If we ranked leagues by number of players from most to least, what percentage of all players do the top two leagues cover together?
س :إذا رتّبنا الدوريات من الأكثر إلى الأقل لاعبين، فما النسبة التي يغطيها أكبر دوريين معاً؟
Tool: Sort leagues by player count descending, then add relative frequencies progressively. Look for whether two leagues alone account for more than 40% of the dataset, revealing concentration in European football.

4- Cross-tabulation Table

Q: Do Forward players have a higher proportion of left-footed players compared to Defenders and Midfielders?
س :هل تضم مجموعة المهاجمين نسبة أعلى من اللاعبين ذوي القدم اليسرى مقارنةً بالمدافعين ولاعبي الوسط؟
Tool: Build a cross-tabulation with Position Group as rows and Preferred Foot (Left / Right) as columns. Look for whether the proportions of left-footed players are consistent across positions — or whether one position shows a higher concentration, which would affect transfer market premiums.

5- Bar Chart
Q: Which league has the highest average Overall Rating among its players?
س: أي دوري يمتلك أعلى متوسط لتقييم الأداء الكلي بين لاعبيه؟
Tool: Bar chart with League on the horizontal axis and mean Overall Rating on the vertical axis. Look for which league tops the chart — and whether the highest-rated league also pays the highest wages.

6- Side-by-Side Bar Chart
Q: Does the Premier League lead in mean Overall Rating across all position groups, or only in some?
س: هل يتصدر الدوري الإنجليزي الممتاز متوسط التقييم الكلي عبر جميع مجموعات المراكز، أم في بعضها فقط؟
Tool: Clustered bar chart with League on the horizontal axis and one bar per Position Group within each cluster. Look for whether one league-position combination clearly dominates, and whether the advantage is consistent across all positions or specific to forwards only.

7- Pie Chart
Q: What share of total market value does each position group contribute to the overall player pool?
س: ما حصة كل مجموعة مراكز من إجمالي القيمة السوقية لجميع اللاعبين؟
Tool: Pie chart with each slice representing a position group's share of total Value_EUR. Look for whether one position group dominates total market value disproportionately compared to its share of player count — Forwards are expected to command a much larger share of value than their numbers suggest.

8- Pareto Chart
Q: Which specific playing positions should a club focus its scouting budget on because they represent the largest share of the player pool?
س: ما المراكز المحددة التي يجب أن تُركّز عليها الأندية ميزانية الكشافة لأنها تمثل الحصة الأكبر من اللاعبين؟
Tool: Bar chart of positions sorted by player count from highest to lowest, with a cumulative percentage line overlaid. Look for the point where the cumulative line crosses 80% — the positions to the left are your vital few that account for the bulk of the professional player market.

9- Histogram
Q: Is the Overall Rating of professional footballers normally distributed, or concentrated in a specific range?
س :هل يتوزع التقييم الكلي للاعبين المحترفين توزيعاً طبيعياً، أم أنه يتمركز في نطاق محدد؟
Tool: Histogram of the Overall column with rating ranges on the horizontal axis and player count on the vertical axis. Look for the peak, the symmetry of the distribution, and whether the shape is bell-shaped — elite players above 85 should appear in a very thin right tail.

10- Scatter Plot
Q: Does a higher Overall Rating consistently lead to a higher market value, and how strong is that pattern?
س :هل يؤدي ارتفاع التقييم الكلي باستمرار إلى ارتفاع القيمة السوقية للاعب، وما قوة هذا النمط؟
Tool: Scatter plot with Overall Rating on the horizontal axis and Value_EUR (or Log_Value) on the vertical axis, with a linear trendline added. Look for direction, the tightness of the cluster around the trendline, and whether the relationship becomes exponential at the top end.

11- Boxplot
Q: Which position group has the most variable market value, and which contains the most extreme superstar outliers?
س :أي مجموعة مراكز تمتلك القيمة السوقية الأكثر تبايناً وتذبذباً، وأيها تحتوي على أكبر عدد من النجوم المتطرفين؟
Tool: Side-by-side boxplot with one box per position group. Look at IQR width (spread of the middle 50%), median position, whisker length, and outlier dots above the upper fence — Forwards are expected to show the widest spread and the most high-value outliers.

12- Mean
Q: On average, what is a professional footballer's weekly wage in the dataset?
س :كم يبلغ متوسط الراتب الأسبوعي للاعب كرة القدم المحترف في قاعدة البيانات؟
Tool: =AVERAGE() on the Wage_EUR column. Check whether the mean is much higher than the median — a large gap signals that a small number of superstar wages are inflating the average and making it unrepresentative of the typical player.

13- Median
Q: What is the market value of the middle-ranked player when all players are sorted by value?
س :كم تبلغ القيمة السوقية للاعب المتوسط في الترتيب عند فرز جميع اللاعبين حسب القيمة؟
Tool: =MEDIAN() on the Value_EUR column. Compare to the mean — because market values are extremely right-skewed, the median is the more honest measure of what a typical professional footballer is worth in the transfer market.

14- Mode
Q: What is the most common Overall Rating score among all players in the dataset?
س :ما التقييم الكلي الأكثر شيوعاً بين جميع اللاعبين في قاعدة البيانات؟
Tool: =MODE() on the Overall column. Identify the most frequently occurring rating value — and check whether it falls near the mean and median or differs significantly, which would reveal an asymmetric distribution.

15- Range
Q: What is the full span of weekly wages from the lowest-paid to the highest-paid player in the dataset?
س :ما المدى الكامل للرواتب الأسبوعية من أقل اللاعبين أجراً إلى أعلاهم في قاعدة البيانات؟
Tool: =MAX() − =MIN() on the Wage_EUR column. Note that the range is extremely sensitive to the single highest-earner — a handful of superstar wages will make the range appear enormous even if 90% of players earn very similarly.

16- Mean Absolute Deviation (MAD)
Q: On average, by how many euros does a typical player's weekly wage deviate from the mean wage — in plain euro terms?
س :باليورو، كم يبلغ متوسط انحراف الراتب الأسبوعي للاعب النموذجي عن المتوسط الحسابي بالقيمة المطلقة؟
Tool: =AVEDEV() on the Wage_EUR column. Look for a large MAD meaning the typical player's wage is routinely far from the average — confirming that the mean is not a reliable description of a 'normal' wage in football.

17- Variance
Q: How do we mathematically quantify the total spread of market values across all players in the dataset?
س :كيف نقيس رياضياً التشتت الكلي للقيم السوقية عبر جميع اللاعبين في قاعدة البيانات؟
Tool: =VAR.S() on the Value_EUR column. Look for an extremely large value —

because variance is in squared euros, it is difficult to interpret directly, but its enormous size confirms that market values are wildly dispersed across the player pool.

18- Standard Deviation

Q: In euros, how much does a typical player's market value differ from the dataset average?

س :باليورو، كم تختلف القيمة السوقية للاعب النموذجي عن متوسط قاعدة البيانات؟

Tool: =STDEV.S() on the Value_EUR column. Look for a standard deviation much larger than the mean — when SD > Mean, it signals extreme inequality in market values and confirms that a few superstars dominate the distribution.

19- Coefficient of Variation (CV)

Q: Which position group has the most inconsistent player quality relative to its average Overall Rating?

س :أي مجموعة مراكز تمتلك أعلى تبايناً في جودة اللاعبين نسبةً إلى متوسط تقييمها الكلي؟

Tool: CV = Standard Deviation ÷ Mean per position group, applied to the Overall Rating column. Look for the highest CV — that position group has the most inconsistent quality, meaning the gap between its best and worst players is largest relative to the group average, making squad planning harder.

20- Skewness

Q: Is the distribution of player market values skewed toward superstar valuations, or toward the lower end of the market?

س :هل يميل توزيع القيم السوقية للاعبين نحو تقييمات النجوم الكبار، أم نحو الطرف الأدنى من السوق؟

Tool: =SKEW() on the Value_EUR column. A strongly positive skewness confirms a long right tail — meaning most players are valued modestly, while a tiny elite (the Mbappés and Haalands) pull the distribution far to the right, creating the statistical fingerprint of a superstar economy.

21- Percentiles & Quartiles

Q: What weekly wage levels define the bottom 25%, the middle range, and the top 10% of earners in the player pool?

س :ما مستويات الراتب الأسبوعي التي تحدد أدنى 25 %والنطاق الأوسط وأعلى 10 %من اللاعبين كسباً؟

Tool: =QUARTILE.INC() for Q1, Q2, Q3 and =PERCENTILE.INC() for the 90th percentile on Wage_EUR. Check how much higher the 90th percentile is compared to Q3 — a very large jump reveals that top earners are in a completely different financial tier from the typical professional.

22- IQR & Outlier Fences
Q: Which players have extreme outlier market values that are statistically unusual compared to the rest of the dataset?
س: أي اللاعبين تُعدّ قيمهم السوقية شاذة ومتطرفة إحصائياً مقارنةً ببقية اللاعبين في قاعدة البيانات؟
Tool: IQR = Q3 − Q1 on Value_EUR. Upper fence = Q3 + 1.5 × IQR. Use =COUNTIF() to count players above the upper fence. These are the statistical superstars — players whose market value is so high that no standard model can explain it from skill data alone.

23- Covariance
Q: Do Overall Rating and Market Value move in the same direction across players in the dataset?
س: هل يتحرك التقييم الكلي والقيمة السوقية في الاتجاه ذاته عبر مختلف اللاعبين في قاعدة البيانات؟
Tool: =COVARIANCE.S() on Overall and Value_EUR. A positive value confirms they move together — higher-rated players tend to have higher values. Note that covariance alone cannot tell us how strong the relationship is; that requires correlation.

24- Correlation
Q: How strong is the linear relationship between a player's Overall Rating and their weekly wage?
س: ما قوة العلاقة الخطية بين التقييم الكلي للاعب وراتبه الأسبوعي؟
Tool: =CORREL() on Overall and Wage_EUR. Interpret sign and magnitude: ≥0.7 strong, 0.4–0.7 moderate, <0.4 weak. Compare to the Overall–Value correlation and note whether Rating predicts Wages or Value more accurately — the answer reveals how the market prices skill.

25- Z-Score
Q: Is a player with an Overall Rating of 85 statistically exceptional, or still within the normal range of professional football?
س: هل يُعدّ لاعب حاصل على تقييم كلي يبلغ 85 درجة استثنائياً إحصائياً، أم لا يزال ضمن النطاق الطبيعي لكرة القدم الاحترافية؟
Tool: Z = (85 − Mean Overall) ÷ Standard Deviation Overall. A Z-score > +2.0 means the player is in the top 2.3% of all professionals — genuinely elite. This is how scouts can objectively quantify how rare a player's rating actually is.

26- Empirical Rule
Q: If Overall Rating were normally distributed, between what two values should approximately 95% of all players fall?
س :لو كان التقييم الكلي يتبع التوزيع الطبيعي، بين أي قيمتين يقع تقريباً 95 %من جميع اللاعبين؟
Tool: Mean ± 2 × Standard Deviation on Overall. Check whether the resulting range is realistic (e.g., does it include negative scores or exceed 99?) — deviations from expected bounds reveal whether the distribution is truly normal or truncated at the top and bottom.

27- Normal Distribution Probability
Q: What proportion of professional players would we expect to be rated 80 or above under a normal distribution assumption?
س :ما النسبة المتوقعة من اللاعبين المحترفين الذين يحصلون على تقييم 80 أو أعلى بافتراض التوزيع الطبيعي؟
Tool: =1 − NORM.DIST(80, mean, stdev, TRUE) to find the proportion above 80. Compare the model prediction to the actual proportion in the data. A large gap confirms the Overall Rating distribution has a thinner elite tail than a normal distribution would predict.

28- Confidence Interval
Q: How confident are we that the sample mean Overall Rating reflects the true average quality of all professional footballers worldwide?
س :ما مدى ثقتنا بأن متوسط التقييم الكلي في العينة يعكس المتوسط الحقيقي لجودة جميع لاعبي كرة القدم المحترفين في العالم؟
Tool: Mean ± CONFIDENCE.NORM(0.05, stdev, n) or CONFIDENCE.T(). Observe how narrow the interval is given the large sample size — with 18,000+ players, the confidence interval should be very tight, demonstrating that the sample provides a highly precise estimate of the true population mean.

29- One-Sample t-Test
Q: Is the average Overall Rating of all players in the dataset significantly greater than 65?
س :هل متوسط التقييم الكلي لجميع اللاعبين في قاعدة البيانات أعلى من 65 درجة بشكل ذي دلالة إحصائية؟
Tool: One-sample t-test comparing the mean Overall to the benchmark of 65. Look for $p < 0.05$ — if the mean is significantly above 65, the dataset confirms that the average professional footballer already clears the basic quality threshold most scouts use.

30- Two-Sample t-Test
Q: Do Premier League players have a significantly higher mean Overall Rating than La Liga players, or is the apparent difference just sampling variation?
س: هل يمتلك لاعبو الدوري الإنجليزي الممتاز متوسط تقييم كلي أعلى بشكل دال إحصائياً من لاعبي الدوري الإسباني؟
Tool: Two-sample t-test (Welch, unequal variances) comparing Overall Rating between Premier League and La Liga players. Check the p-value and the practical difference — even if statistically significant with large samples, a difference of less than 1 Overall point has no real scouting importance.

31- Paired t-Test
Q: Is a player's Potential rating significantly higher than their current Overall Rating on average across the full dataset?
س: هل التقييم المستقبلي المتوقع للاعب أعلى بشكل دال إحصائياً من تقييمه الكلي الحالي في المتوسط عبر قاعدة البيانات؟
Tool: Create a Gap column = Potential − Overall, then run a one-sample t-test on Gap vs 0. A positive, statistically significant mean gap confirms that the typical professional has unrealised development headroom — the statistical basis for buying young talent at a discount to their future value.

32- One-Way ANOVA
Q: Do the four position groups (Forwards, Midfielders, Defenders, Goalkeepers) differ significantly in their mean Pace attribute?
س: هل تختلف مجموعات المراكز الأربع (مهاجمون، وسط، مدافعون، حراس) اختلافاً جوهرياً في متوسط سرعة اللاعبين؟
Tool: ANOVA (Single Factor) with Pace as the dependent variable and Position Group as the grouping factor. If $p < 0.05$, at least one group differs — then run post-hoc tests (Tukey HSD) to identify which pairs are different. The result determines whether Pace benchmarks must be position-specific in scouting.

33- Chi-Square Test
Q: Is there a statistically significant relationship between a player's preferred foot and their position group?
س: هل توجد علاقة ذات دلالة إحصائية بين القدم المفضلة للاعب ومجموعة مركزه في الملعب؟
Tool: =CHISQ.TEST() on the cross-tabulation of Preferred Foot (Left / Right) by Position Group. $p < 0.05$ would indicate that foot preference is not randomly distributed across positions — but if $p > 0.05$, the left-footed premium in the transfer market reflects overall scarcity, not positional clustering.

34- Simple Linear Regression
Q: How much does a player's market value increase for each 1-point improvement in their Overall Rating?
س :كم تزداد القيمة السوقية للاعب لكل زيادة مقدارها درجة واحدة في تقييمه الكلي؟
Tool: Regression with Log(Value_EUR) as the dependent variable and Overall Rating as the predictor. Interpret the coefficient as a percentage change in value per point (log-linear model), the R^2 showing how much of value variation rating explains, and the p-value confirming the relationship is statistically real and not due to chance.

35- Multiple Linear Regression
Q: How well can we predict a player's market value using Overall Rating, Age, and Wage together — and which of these three factors matters most?
س :ما دقة التنبؤ بالقيمة السوقية للاعب باستخدام التقييم الكلي والعمر والراتب معاً — وأيها الأكثر تأثيراً؟
Tool: Multiple regression with Log(Value_EUR) as the dependent variable and Overall Rating, Age, and Wage_EUR as predictors. Compare individual coefficients to see which has the strongest effect, check whether R^2 improves meaningfully over the simple model, and assess whether Age has a negative coefficient — confirming that the market discounts older players even when skill is held constant.